

✓ Setup Complete - Mobile CPR Detector

Yang Sudah Dilakukan

1. ✓ Node.js Upgraded

- **Sebelum:** Node v20.18.3
- **Sekarang:** Node v24.12.0 (LTS terbaru)
- **NPM:** v11.6.2

2. ✓ Expo SDK Updated

- Menggunakan **Expo SDK 54** (terbaru)
- Compatible dengan Expo Go app versi terbaru di smartphone

3. ✓ Dependencies Fixed

Package yang sudah diupdate:

```
{
  "expo": "~54.0.0",
  "expo-camera": "~17.0.10",
  "expo-status-bar": "~3.0.9",
  "react": "19.1.0",
  "react-native": "0.81.5",
  "react-native-svg": "15.12.1",
  "@tensorflow/tfjs": "^4.11.0",
  "@tensorflow-models/pose-detection": "^2.1.0"
}
```

4. ✓ Camera API Updated

- Menggunakan **CameraView** dan **useCameraPermissions** (API terbaru)
- Compatible dengan expo-camera v17

5. ✓ Assets Added

- ✓ icon.png
- ✓ splash.png
- ✓ adaptive-icon.png
- ✓ favicon.png
- ✓ models/ folder

Cara Menjalankan

1. Install Dependencies (jika belum)

```
cd /home/amelia-ocha/Documents/project_izin/mobile-cpr-detector
npm install
```

2. Start Expo Dev Server

```
npx expo start -c
```

3. Scan QR Code

- **Android:** Buka app **Expo Go**, klik "Scan QR code"
- **iOS:** Buka app **Camera**, arahkan ke QR code

4. Grant Permission

Saat app terbuka pertama kali:

1. App akan meminta **Camera Permission** ✓
2. Klik **"Allow"** atau **"Grant Permission"**
3. Kamera akan langsung terbuka! 📷



Expected Behavior

Saat app berhasil jalan:

1. Loading Screen

- "Loading AI Model..."
- "Initializing MoveNet Pose Detection"

2. Permission Screen (jika belum granted)

- "X Camera permission required"
- Button "Grant Permission"

3. Camera Screen ✓

- Live camera feed terlihat
- Top bar: FPS counter
- Status: "⚠️ Waiting for pose..." atau "CPR DETECTED"
- Bottom controls: "👁️ Show Skeleton" button
- Instructions box di kiri atas



Troubleshooting

Masalah: Kamera Masih Hitam

Solusi 1: Force Restart

```
# Stop server (Ctrl+C)
# Clear cache
npx expo start -c
```

Solusi 2: Reinstall di Device

1. Di Expo Go app, long-press pada project
2. Delete/Remove project
3. Scan QR code lagi

Solusi 3: Check Permission di Device

- **Android:**
 - Settings → Apps → Expo Go → Permissions → Camera ✓
- **iOS:**
 - Settings → Expo Go → Camera ✓

Masalah: "SDK Version Mismatch"

Solusi: Update Expo Go app di smartphone

1. Buka Play Store (Android) atau App Store (iOS)
2. Search "Expo Go"
3. Update ke versi terbaru

Masalah: Metro Bundler Error

Solusi: Clean install

```
rm -rf node_modules package-lock.json .expo
npm install
npx expo start -c
```

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III Performance Notes

Current Limitations:

1. **Frame processing** masih simplified (placeholder)
2. **TensorFlow.js di React Native** memiliki keterbatasan performa
3. Untuk **real pose detection**, perlu implementasi native bridge

Recommended Next Steps:

1. Implement actual camera frame capture
2. Optimize TensorFlow.js backend (webgl/wasm)
3. Consider using **TensorFlow Lite** untuk performa lebih baik

4. Add frame rate throttling (currently placeholder)

Features yang Sudah Ada

- ✓ Camera view dengan live feed
- ✓ Permission handling
- ✓ UI Overlay (CPR point, skeleton toggle)
- ✓ FPS counter
- ✓ Status indicator
- ✓ Instructions overlay
- ✓ Clean, modern UI

SOON Next Development Tasks

1. Implement actual pose detection

- Connect camera frames to TensorFlow.js
- Process frames at ~5-10 FPS
- Display detected keypoints

2. Add CPR calculation

- Calculate compression point from detected pose
- Visualize with marker overlay
- Show confidence score

3. Performance optimization

- Optimize tensor operations
- Add frame skipping
- Memory management

4. Testing

- Test on different devices
- Test different lighting conditions
- Test different body positions

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Tech Stack

- **Framework:** React Native + Expo SDK 54
- **Camera:** expo-camera v17
- **AI/ML:** TensorFlow.js + MoveNet Pose Detection
- **Graphics:** react-native-svg
- **Runtime:** Node.js v24.12.0

Learning Resources

- [Expo Camera Docs](#)
- [TensorFlow.js](#)

- [MoveNet Model](#)
 - [React Native](#)
-

☆☆ Success Checklist

Jika semuanya berhasil, Anda akan lihat:

- ☐ QR code muncul di terminal
- ☐ App terbuka di smartphone
- ☐ Loading screen dengan "Loading AI Model..."
- ☐ Permission request muncul
- ☐ Setelah allow: **KAMERA TERBUKA** ✓
- ☐ UI overlay terlihat (FPS, status, buttons)
- ☐ Console log menampilkan:

```
TensorFlow.js ready  
Backend: webgl  
MoveNet detector loaded
```

Selamat! 🎉 App sudah berjalan dengan baik!

Last updated: 26 December 2025

Node version: v24.12.0

Expo SDK: 54